3.3 Topic 1: Energy Security

Energy	Humans use a wide range of energy sources. However, they currently depend heavily on fossil fuels; ultimately a finite resource. Physical factors mean that the geography of fossil fuels, and renewable energy potential, is uneven. Access to energy resources partly depends on physical factors, but also on the availability of capital and technology. Combined, these factors result in some areas experiencing energy surpluses (energy security), while other areas suffer energy deficit (energy insecurity). Economic wealth and potential depend on energy supply, and with demand for energy growing, there is potential for conflict over supply.
Security	Securing supply is a key issue, and there are potential environmental and political risks associated with exploiting new resources. Major players in the energy issue, such as TNCs and IGOs, are powerful and their role is increasingly important. The future of energy exploitation and supply is unclear. This is partly due to uncertainty about how long fossil fuel reserves will last, and partly due to the difficulties of finding acceptable and cost effective alternative energy sources. There is a wide range of potential future energy scenarios, each with its own supporters.

1 Energy supply, demand and security

Enquiry question: *To what extent is the world 'energy secure' at present?*

What students need to learn	Suggested teaching and learning
■ There are many energy sources that can be classified in different ways (flows of renewable sources, stocks of non-renewable and recyclable sources) and that have different environmental costs.	■ Investigating types of energy resources, their classification, and contrasting the environmental impacts associated with their production and use.
■ Access to and consumption of energy resources, both renewable and non-renewable, is not evenly distributed, and depends on physical factors, cost, technology and public perception. Some areas suffer from energy poverty, while others have a surplus.	■ Examining the distribution of fossil fuel resources, and renewable potential, globally and in contrasting countries.

■ Demand for energy is growing globally, and at regional and local scales, especially in developed and emergent economies such as China and India.	■ Examining trends in global energy supply and demand by source, type of economy and economic sector.
■ Energy security depends on resource availability (domestic and foreign) and security of supply, which can be affected by geopolitics, and is a key issue for many economies.	■ Developing an awareness that that there is little excess capacity to ease pressure on energy resources and therefore energy insecurity is rising, particularly for finite resources.

2 The impacts of energy insecurity

Enquiry question: *What are the potential impacts of an increasingly 'energy insecure' world?*

What students need to learn	Suggested teaching and learning
■ Energy pathways, between producers and consumers, are complex and show increasing levels of risk eg the trans-Siberian gas pipeline into Western Europe, or Middle Eastern supplies.	■ Examining developments in the geography of energy infrastructure and supply pathways that connect producers to consumers.
■ There are real risks, in economic and political terms, if energy supplies are disrupted.	■ Developing awareness that tensions exist between energy producers and consumers, and that these can result in increased risk (rising costs) and conflict.
■ Increasing energy insecurity has stimulated exploration of technically difficult and environmentally sensitive areas, such as the Arctic circle, the West Shetland field and Canadian oil shales, which may incur environmental costs.	■ Investigating the costs and benefits of exploiting new areas and resources, in economic, human and environmental terms.
■ Energy TNCs, OPEC countries and other large producers are increasingly powerful players in the global supply of energy.	■ Investigating the increasing economic and political power of selected energy TNCs and producer groups.